SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 5 (TOPIC 5)

Theme: MICROCOMPUTER APPLICATIONS: WORD PROCESSING, ADVANCED SPREADSHEETS & INTERNET PROTOCOLS

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 5.1 (Word Processors & Formatting), Subtopic 5.2 (Spreadsheets & Formulas), Subtopic 5.3 (Web Browsing & Email Protocols)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 10 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Show all intermediate formulas, binary conversions, architectural diagrams, command syntax, and arithmetic calculations clearly.
4. Where design questions require software configurations, provide exact command line syntax (Bash/DOS) or functional formulas (Excel) as specified.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL SOFTWARE CONCEPTS [25 MARKS]

Question 1 [5 Marks]: In word processing systems (e.g., Microsoft Word, LibreOffice Writer), differentiate between a 'Section Break (Next Page)' and a simple 'Page Break'. Describe a specific scenario in compiling an academic undergraduate thesis where a Section Break is mandatory.

Question 2 [5 Marks]: Explain the mechanical architecture of Mail Merge. Identify the three principal components involved in any mail merge pipeline and explain how conditional fields (`IF...THEN...ELSE`) enable dynamic document customization.

Question 3 [5 Marks]: In spreadsheet modeling (e.g., Microsoft Excel), explain the fundamental distinction between: (a) Relative Cell Referencing (`A1`), (b) Absolute Cell Referencing (`$A$1`), and (c) Mixed Cell Referencing (`$A1` and `A$1`). Give a mathematical formula example illustrating where mixed referencing is required.

Question 4 [5 Marks]: Explain the architectural difference between the `VLOOKUP` function and the modern `INDEX` + `MATCH` combination in Excel. Detail two major structural limitations of `VLOOKUP` that `INDEX-MATCH` and `XLOOKUP` eliminate.

Question 5 [5 Marks]: Contrast the two primary electronic mail retrieval protocols: POP3 (Post Office Protocol v3) and IMAP4 (Internet Message Access Protocol v4). Under what operational conditions is IMAP4 strictly required for university staff and students?

SECTION B: ADVANCED SPREADSHEET ALGORITHMS & PROTOCOL ANALYSIS [35 MARKS]

Question 6 [15 Marks]: Spreadsheet Mathematical & Statistical Modeling:
The SUA Farm Manager records monthly milk production data for 200 dairy cows in an Excel worksheet `MilkYield`. The columns are arranged as follows: Column A = Cow ID (`C001` to `C200`), Column B = Breed (`Friesian`, `Ayrshire`, `Boran`), Column C = Age in Years, Column D = Daily Yield (Liters), Column E = Feed Cost (TZS), Column F = Health Status (`Healthy`, `Treated`, `Quarantine`).
Write the exact, syntactically correct Excel formulas to compute each of the following: [3 Marks each]
(a) The total milk yield produced exclusively by `Friesian` cows.
(b) The average daily milk yield of cows that are both `Ayrshire` breed AND have a Health Status of `Healthy`.
(c) The count of cows whose Age is greater than or equal to 4 years AND whose daily yield exceeds 18.5 Liters.
(d) A formula in Column G that categorizes daily yield: If Yield ≥ 20 Liters, display 'High Producer'; if Yield is between 12 and 19.9 Liters, display 'Moderate Producer'; otherwise display 'Low Producer - Cull Review'.
(e) A lookup formula that searches for a user-input Cow ID in cell `J2` and returns that cow's Breed and Feed Cost using `INDEX` and `MATCH`.

Question 7 [10 Marks]: Financial Spreadsheet Modeling: Capital Budgeting for Agro-Processing Equipment:
SUA Agricultural Investment Directorate is evaluating the purchase of a commercial solar grain dryer costing 45,000,000 TZS. The machine generates the following net cash inflows over a 5-year operational lifecycle: Year 1 = 12,000,000 TZS; Year 2 = 14,500,000 TZS; Year 3 = 16,000,000 TZS; Year 4 = 15,000,000 TZS; Year 5 = 11,000,000 TZS. The university discount rate is 10.5% per annum.
(a) Formulate the mathematical equation for Net Present Value (NPV). [3 Marks]
(b) Write the exact Excel formula using the `NPV` function to calculate the Net Present Value of this investment. [4 Marks]
(c) Explain how the `IRR` (Internal Rate of Return) function is used in conjunction with the discount rate to make an accept/reject investment decision. [3 Marks]

Question 8 [10 Marks]: Web Protocols & HTTP/HTTPS Security Cycle:
(a) Dissect the components of the Uniform Resource Locator (URL): `https://www.sua.ac.tz:443/academics/courses?dept=diit&term;=2#syllabus`. Identify: Protocol/Scheme, Fully Qualified Domain Name (FQDN), Port, Path, Query String, and Anchor/Fragment. [4 Marks]
(b) Explain the HTTP Request-Response cycle. Differentiate between the HTTP `GET` and `POST` methods regarding parameter transmission and security. [3 Marks]
(c) Interpret the following HTTP response status codes: 200 OK, 301 Moved Permanently, 401 Unauthorized, 404 Not Found, 500 Internal Server Error, and 503 Service Unavailable. [3 Marks]

SECTION C: APPLIED DOCUMENT & ENTERPRISE ARCHITECTURE [40 MARKS]

Question 9 [15 Marks]: Academic Document Engineering & Master Dissertation Compilation:
A postgraduate student at SUA is formatting a 120-page dissertation in Microsoft Word / LibreOffice Writer requiring strict university formatting standards:
- Preliminary pages (Title, Declaration, Abstract, Table of Contents) must use Roman numerals (i, ii, iii, iv) centered at the bottom.
- Chapter 1 onwards must begin with Arabic numeral '1' at the top-right corner.
- Chapter 4 contains a wide 8-column agricultural field trial table that must be displayed in Landscape orientation, while preceding and succeeding pages remain Portrait.
(a) Provide step-by-step technical instructions detailing how to insert Section Breaks, unlink headers and footers ('Link to Previous'), configure page numbering formats, and isolate page orientation for Chapter 4 without corrupting the rest of the document. [8 Marks]
(b) Explain how the automated Table of Contents (TOC) and Table of Figures are constructed using Heading Styles (Heading 1, Heading 2) and Caption tags. Why should a professional academic document never contain manually typed tables of contents? [4 Marks]
(c) Explain the utility of the 'Track Changes' feature and document compare tools in collaborative academic review between students and thesis supervisors. [3 Marks]

Question 10 [15 Marks]: Electronic Mail Infrastructure & Anti-Spoofing Architecture:
The SUA ICT Directorate operates the university mail server domain `@sua.ac.tz`. Recently, fraudulent emails claiming to be from the Vice Chancellor's office were received by students.
(a) Trace the complete path of an email sent via SMTP from `instructor@sua.ac.tz` to `student@gmail.com`. Detail the interactions between Mail User Agent (MUA), Mail Submission Agent (MSA), Mail Transfer Agent (MTA), and Mail Delivery Agent (MDA). [6 Marks]
(b) Explain how the three foundational email authentication protocols protect against domain spoofing and phishing: [6 Marks]
1. SPF (Sender Policy Framework): How DNS TXT records specify authorized sending mail servers.
2. DKIM (DomainKeys Identified Mail): How public-key digital signatures in message headers verify message integrity.
3. DMARC (Domain-based Message Authentication, Reporting, and Conformance): How policy enforcement and reporting align SPF and DKIM results.
(c) Construct a valid DNS TXT record for SPF that authorizes the mail server with IP `196.216.180.15` to send email for `sua.ac.tz` and instructs receiving servers to hard-fail (`-all`) all other sources. [3 Marks]

Question 11 [10 Marks]: Pivot Table Business Intelligence for Agricultural Cooperatives:
A Morogoro sugarcane cooperative collects harvest deliveries from 1,500 outgrowers across 6 geographical zones. The transactional ledger contains 25,000 rows with: Delivery ID, Date, Farmer ID, Zone, Tons Delivered, Sugar Content (Brix %), and Payment Status.
(a) Explain how an Excel Pivot Table can be constructed to summarize total tonnage and average Brix percentage per agricultural zone. Specify which fields are dragged to: Rows, Columns, Values, and Filters. [5 Marks]
(b) Describe how to insert a Slicer and a PivotChart to create an interactive management dashboard for the cooperative board of directors. [3 Marks]
(c) What is a 'Calculated Field' in a Pivot Table? Give an example formula to calculate Gross Farmer Payment based on `Tons Delivered * 85000 TZS * (Brix / 10)`. [2 Marks]